

Qualification Pack



Process Safety Supervisor Oil Gas

QP Code: HYC/Q3603

Version: 1.0

NSQF Level: 5

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Qualification Pack

Contents

HYC/Q3603: Process Safety Supervisor Oil Gas	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
HYC/N3606: Develop policies, SOPs and train to ensure safety at workplace	5
HYC/N3607: Incorporate the PSM system into the culture of leadership and organization	11
HYC/N3608: Implement PSM practices within the organization	17
HYC/N3609: Carry out PSM in oil and gas industries	28
HYC/N9301: Working effectively in a team	36
HYC/N9302: Maintain health, safety and security procedures	40
DGT/VSQ/N0102: Employability Skills (60 Hours)	45
Assessment Guidelines and Weightage	52
<i>Assessment Guidelines</i>	52
<i>Assessment Weightage</i>	53
Acronyms	54
Glossary	56

Qualification Pack

HYC/Q3603: Process Safety Supervisor Oil Gas

Brief Job Description

The process safety officer is responsible for conducting regular site inspections to identify potential hazards and ensure that safety measures are in place and properly maintained, investigate accidents and incidents to determine the root causes, and recommend corrective actions to prevent future occurrences.

Personal Attributes

The individual should have a good knowledge and communication skills on process envelopes including safety aspects. The person must have supervisory skills. The individual should be proactive, disciplined, safety-conscious, courageous and work individually and in a team.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [HYC/N3606: Develop policies, SOPs and train to ensure safety at workplace](#)
2. [HYC/N3607: Incorporate the PSM system into the culture of leadership and organization](#)
3. [HYC/N3608: Implement PSM practices within the organization](#)
4. [HYC/N3609: Carry out PSM in oil and gas industries](#)
5. [HYC/N9301: Working effectively in a team](#)
6. [HYC/N9302: Maintain health, safety and security procedures](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Hydrocarbon
Sub-Sector	Upstream, Midstream, Downstream
Occupation	Management of Health, Safety and Environment (HSE)
Country	India
NSQF Level	5

Qualification Pack

Credits	15.6
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2141.2500
Minimum Educational Qualification & Experience	Completed 3 year UG degree (Science) with 3 Years of experience OR Completed 3 year UG degree ((Engineering) BE/BTech) with 1-2 Years of experience
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	30/05/2027
NSQC Approval Date	30/05/2024
Version	1.0
Reference code on NQR	QG-05-HY-02620-2024-V1-HSSC
NQR Version	1.0

Qualification Pack

HYC/N3606: Develop policies, SOPs and train to ensure safety at workplace

Description

This unit covers Design and implement workplace safety policies, SOPs, and training programs to ensure a safe work environment, promoting employee well-being and accident prevention.

Scope

The scope covers the following :

- This unit/task covers the following:
- stakeholders
- Training & competency development

Elements and Performance Criteria

stakeholders

To be competent, the user/individual on the job must be able to:

- PC1.** Develop and implement safety policies and procedures to ensure compliance with industry standards and regulations
- PC2.** Implement policies and procedures that ensure consistent service delivery that is safe and efficient.
- PC3.** Follow Oil Industry Safety Directorate (OISD) standards related to process safety, system, equipment and appliances in oil and gas industry.
- PC4.** Prepare safety documents such as MSDS, Safety manual, training record, etc.
- PC5.** Form a safety committee in the organization
- PC6.** Collaborate with senior leadership to set organizational health & safety goals for the company.
- PC7.** Puts safety first in all company transactions following Safety policies and ensuring others comply to protect people, environment, assets, and the company's reputation in company operations.
- PC8.** Coordinate with management, regulatory agencies, and other stakeholders to address safety concerns and improve overall safety performance.
- PC9.** Convey process information like P&ID, SOP, MOC, SIMOPS

Training & competency development

To be competent, the user/individual on the job must be able to:

- PC10.** Identify the safety training requirement
- PC11.** Create a comprehensive matrix outlining initial, mandatory, refresher, and advanced training needs, along with evaluation techniques
- PC12.** Provide safety training for employees, including hazard recognition, safe work practices, and emergency response procedures.
- PC13.** Liaise with team members in order to create need assessments and gap analyses to identify opportunities for various types of safety-related programs.

Qualification Pack

- PC14.** Ensure all the personnel wear appropriate personal protective equipment (PPE) and follow established safety protocols
- PC15.** Demonstration of API RP 754 (process safety performance indicator)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand how to make and put in place safety rules to follow industry standards and laws, including knowing the important legal rules and best ways to do things.
- KU2.** able to set up rules and ways of doing things that make sure services are always safe and work well. This means making safety a part of everything we do.
- KU3.** Knowing and following the safety rules set by the Oil Industry Safety Directorate (OISD) for processes, systems, equipment, and appliances in the oil and gas industry to keep the workplace safe.
- KU4.** Being good at making different safety papers like Material Safety Data Sheets (MSDS), safety guides, and training records. This helps to accurately write down safety rules and procedures.
- KU5.** Knowing how to set up and run a safety team in the organization. This includes deciding who's in it, what they're responsible for, and what they need to achieve to keep safety standards high.
- KU6.** Being good at working together with senior leaders to decide on health and safety goals for the company. This means making sure these goals match up with what the company wants to achieve overall
- KU7.** Making sure safety is the most important thing in all company activities. This includes following safety rules, making others follow them too, and keeping everyone safe, protecting the environment, company assets, and its reputation
- KU8.** able to work with managers, government agencies, and other involved parties to deal with safety issues and make safety better overall. This means everyone working together to make safety improvements happen.
- KU9.** Understanding how to identify the specific safety training needs based on job roles, tasks, and regulatory requirements within the organization
- KU10.** Understanding how to create a detailed list of all the safety training needed.
- KU11.** Ability to provide safety training for employees, covering topics such as hazard recognition, safe work practices, emergency response procedures, and other relevant safety protocols.
- KU12.** able to work together with team members to figure out what safety improvements are needed.
- KU13.** Knowing how important it is for everyone to wear the proper protective gear and follow safety rules.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** keep abreast by reading about new policies at an organization level
- GS2.** communicate effectively in writing

Qualification Pack

- GS3.** convey information effectively
- GS4.** discuss observations, new policies and schedules and activities
- GS5.** listen with full attention and comprehend the information given by various source about the site or equipment
- GS6.** use reasoning skills to identify and resolve problems
- GS7.** organize all process manuals so that sorting/ accessing information is easy
- GS8.** analyse information and use logic to address safety-related issues and problems.
- GS9.** determine the safety measures to be followed and how changes in conditions operations, and the environment will affect outcomes.
- GS10.** undertake and express new ideas and initiatives
- GS11.** Supervise periodically all the process safety systems

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>stakeholders</i>	12	18	-	-
PC1. Develop and implement safety policies and procedures to ensure compliance with industry standards and regulations	1	2	-	-
PC2. Implement policies and procedures that ensure consistent service delivery that is safe and efficient.	1	2	-	-
PC3. Follow Oil Industry Safety Directorate (OISD) standards related to process safety, system, equipment and appliances in oil and gas industry.	1	2	-	-
PC4. Prepare safety documents such as MSDS, Safety manual, training record, etc.	2	2	-	-
PC5. Form a safety committee in the organization	1	2	-	-
PC6. Collaborate with senior leadership to set organizational health & safety goals for the company.	1	2	-	-
PC7. Puts safety first in all company transactions following Safety policies and ensuring others comply to protect people, environment, assets, and the company's reputation in company operations.	2	2	-	-
PC8. Coordinate with management, regulatory agencies, and other stakeholders to address safety concerns and improve overall safety performance.	1	2	-	-
PC9. Convey process information like P&ID, SOP, MOC, SIMOPS	2	2	-	-
<i>Training & competency development</i>	8	12	-	-
PC10. Identify the safety training requirement	2	2	-	-
PC11. Create a comprehensive matrix outlining initial, mandatory, refresher, and advanced training needs, along with evaluation techniques	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Provide safety training for employees, including hazard recognition, safe work practices, and emergency response procedures.	2	2	-	-
PC13. Liaise with team members in order to create need assessments and gap analyses to identify opportunities for various types of safety-related programs.	1	2	-	-
PC14. Ensure all the personnel wear appropriate personal protective equipment (PPE) and follow established safety protocols	1	2	-	-
PC15. Demonstration of API RP 754 (process safety performance indicator)	1	2	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N3606
NOS Name	Develop policies, SOPs and train to ensure safety at workplace
Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Management of Health, Safety and Environment (HSE)
NSQF Level	5
Credits	3.5
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

HYC/N3607: Incorporate the PSM system into the culture of leadership and organization

Description

This unit covers Integrate Process Safety Management (PSM) principles into organizational leadership, fostering a culture where safety is paramount, ensuring compliance, risk mitigation, and continuous improvement

Scope

The scope covers the following :

- This unit/task covers the following:
- Leadership & Organization
- Management of Change (MOC)
- Emergency Planning & Response
- Incident Management
- Audit Protocol
- Safety Performance Evaluation
- Asset Integrity Management (AIM)

Elements and Performance Criteria

Leadership & Organization

To be competent, the user/individual on the job must be able to:

- PC1.** Emphasize leadership within process units, and provide awareness on hazard and risk among the leadership team
- PC2.** Introduce organizational learning principles and establish clear process safety responsibilities
- PC3.** Manage worker engagement and participation effectively.
- PC4.** Raise hazard and risk awareness among the leadership team.
- PC5.** Implement safety policies, committees, and promotion activities.

Management of Change (MOC)

To be competent, the user/individual on the job must be able to:

- PC6.** Establish a process for approving equipment, process, method, and personnel changes
- PC7.** Provide training to all relevant parties regarding the impact of these changes
- PC8.** Maintain proper documentation for all approved changes

Emergency Planning & Response

To be competent, the user/individual on the job must be able to:

- PC9.** Appreciate the role in crisis management
- PC10.** Create a comprehensive emergency response plan
- PC11.** Train the response team, include essential information

Incident Management

To be competent, the user/individual on the job must be able to:

Qualification Pack

PC12. Recognize process safety-related incidents

PC13. Utilize incident reporting matrix for documentation

PC14. Investigate incidents, analyze root causes, and address near misses

Audit Protocol.

To be competent, the user/individual on the job must be able to:

PC15. conduct Internal Safety Audits as per the international standards

PC16. Check process and documents related to Incident Management, Work Permit System, Emergency Response Plans, Safety Audits, Bulk Storage Operations, etc. as applicable

PC17. Implement mechanisms to address external audit recommendations

Safety performance evaluation.

To be competent, the user/individual on the job must be able to:

PC18. monitor historical safety data like

PC19. track lagging Indicators and assess leading Indicators

PC20. evaluate proactive safety measures

Asset Integrity Management (AIM)

To be competent, the user/individual on the job must be able to:

PC21. Prevent adverse outcomes due to compromised asset integrity.

PC22. Evaluate equipment options based on operating conditions.

PC23. Isolate chemical or process lines using inline valves and drain/vent lines.

PC24. Develop maintenance plans guided by risk assessments.

PC25. Demonstration of SOE(Safe Operating Envelope) & IOW (Integrity Operating Windows)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Ability to lead and organize teams effectively in various safety and operational scenarios.

KU2. Understanding of organizational structures and processes to ensure efficient safety management

KU3. Knowledge of procedures and protocols for managing changes in processes, equipment, or procedures to maintain safety and compliance.

KU4. Understanding of emergency response protocols, including evacuation procedures, communication plans, and coordination with emergency services.

KU5. Ability to manage and respond to incidents effectively, including investigation, reporting, and corrective action implementation

KU6. Knowledge of audit procedures and protocols for assessing compliance with safety standards and regulations

KU7. Understanding of methods and metrics used to evaluate safety performance and identify areas for improvement.

KU8. Knowledge of strategies and practices for ensuring the integrity and reliability of assets, including equipment and infrastructure.

KU9. Ability to assess risks associated with operational activities and implement appropriate control measures

Qualification Pack

- KU10.** Understanding of relevant safety regulations and standards governing operations in the industry.
- KU11.** Knowledge of training programs and initiatives aimed at enhancing safety awareness and competency among employees.
- KU12.** Effective communication with stakeholders, employees, and regulatory authorities regarding safety matters.
- KU13.** Ability to manage crisis situations effectively, including communication, decision-making, and resource allocation
- KU14.** Understanding of quality management systems and their application to safety processes and procedures.
- KU15.** Ability to monitor safety performance indicators and take corrective actions as necessary.
- KU16.** Knowledge of procedures for managing contractors and ensuring their compliance with safety requirements
- KU17.** Ability to conduct root cause analysis to identify underlying factors contributing to safety incidents or near misses.
- KU18.** Commitment to ongoing improvement of safety processes and procedures based on lessons learned and best practices
- KU19.** Understanding of environmental considerations and regulations relevant to operations in the industry.
- KU20.** Knowledge of budgeting and resource allocation processes to support safety initiatives.
- KU21.** Awareness of legal and ethical considerations relevant to safety management in the industry.
- KU22.** Familiarity with industry best practices and benchmarks for safety management and performance

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** keep abreast by reading about new policies at an organization level
- GS2.** communicate effectively in writing
- GS3.** convey information effectively
- GS4.** discuss observations, new policies and schedules and activities
- GS5.** listen with full attention and comprehend the information given by various source about the site or equipment
- GS6.** use reasoning skills to identify and resolve problems
- GS7.** organize all process manuals so that sorting/ accessing information is easy
- GS8.** analyse information and use logic to address safety-related issues and problems.
- GS9.** determine the safety measures to be followed and how changes in conditions operations, and the environment will affect outcomes.
- GS10.** undertake and express new ideas and initiatives
- GS11.** Supervise periodically all the process safety systems

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Leadership & Organization</i>	5	10	-	-
PC1. Emphasize leadership within process units, and provide awareness on hazard and risk among the leadership team	1	2	-	-
PC2. Introduce organizational learning principles and establish clear process safety responsibilities	1	2	-	-
PC3. Manage worker engagement and participation effectively.	1	2	-	-
PC4. Raise hazard and risk awareness among the leadership team.	1	2	-	-
PC5. Implement safety policies, committees, and promotion activities.	1	2	-	-
<i>Management of Change (MOC)</i>	3	6	-	-
PC6. Establish a process for approving equipment, process, method, and personnel changes	1	2	-	-
PC7. Provide training to all relevant parties regarding the impact of these changes	1	2	-	-
PC8. Maintain proper documentation for all approved changes	1	2	-	-
<i>Emergency Planning & Response</i>	3	6	-	-
PC9. Appreciate the role in crisis management	1	2	-	-
PC10. Create a comprehensive emergency response plan	1	2	-	-
PC11. Train the response team, include essential information	1	2	-	-
<i>Incident Management</i>	3	6	-	-
PC12. Recognize process safety-related incidents	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Utilize incident reporting matrix for documentation	1	2	-	-
PC14. Investigate incidents, analyze root causes, and address near misses	1	2	-	-
<i>Audit Protocol.</i>	3	6	-	-
PC15. conduct Internal Safety Audits as per the international standards	1	2	-	-
PC16. Check process and documents related to Incident Management, Work Permit System, Emergency Response Plans, Safety Audits, Bulk Storage Operations, etc. as applicable	1	2	-	-
PC17. Implement mechanisms to address external audit recommendations	1	2	-	-
<i>Safety performance evaluation.</i>	3	6	-	-
PC18. monitor historical safety data like	1	2	-	-
PC19. track lagging Indicators and assess leading Indicators	1	2	-	-
PC20. evaluate proactive safety measures	1	2	-	-
<i>Asset Integrity Management (AIM)</i>	5	10	-	-
PC21. Prevent adverse outcomes due to compromised asset integrity.	1	2	-	-
PC22. Evaluate equipment options based on operating conditions.	1	2	-	-
PC23. Isolate chemical or process lines using inline valves and drain/vent lines.	1	2	-	-
PC24. Develop maintenance plans guided by risk assessments.	1	2	-	-
PC25. Demonstration of SOE(Safe Operating Envelope) & IOW (Integrity Operating Windows)	1	2	-	-
NOS Total	25	50	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N3607
NOS Name	Incorporate the PSM system into the culture of leadership and organization
Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Management of Health, Safety and Environment (HSE)
NSQF Level	5
Credits	2.5
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

HYC/N3608: Implement PSM practices within the organization

Description

This unit covers basic practices that Execute Process Safety Management (PSM) protocols within the organization, embedding safety procedures, risk assessments, and regulatory compliance to safeguard personnel, assets, and the environment

Scope

The scope covers the following :

- This unit/task covers the following:
- Process Safety Standards
- Process Safety Information
- Hazard Identification & Risk Analysis (HIRA)
- Operating Procedures
- Pre-Start-Up, Start-Up, and Shut-Down Review
- Work Permit System
- Bulk Storage Operations
- Electrical Safety
- Fire Safety
- Dangerous Substances
- Reaction Hazards
- Safety in Using Steam
- Dust Explosions

Elements and Performance Criteria

Process Safety standards

To be competent, the user/individual on the job must be able to:

- PC1.** comprehend the Oil Industry Safety Directorate (OISD) standards pertaining to process safety, systems, equipment, and appliances in the oil and gas industry

Process Safety Information.

To be competent, the user/individual on the job must be able to:

- PC2.** Review safety documents such as MSDS, Safety Manual, and Training records.
- PC3.** Examine design aspects including interlock systems, alarm systems, and electrical classification.
- PC4.** Check P&ID, SOPs, MOC, and SIMOPS.

Hazard Identification & Risk Analysis (HIRA)

To be competent, the user/individual on the job must be able to:

- PC5.** Analyze risks and grasp the purpose and utility of risk analysis
- PC6.** Evaluate both quantitative and qualitative aspects in Quantitative Risk Assessment (QRA) studies
- PC7.** Conduct HAZOP studies to identify potential hazards

Qualification Pack

PC8. Implement strategies for effective barrier management

PC9. Learn concepts like HAZID, FMEA, and ALARP (As Low As Reasonably Practicable)

Operating Procedures.

To be competent, the user/individual on the job must be able to:

PC10. Recognize the boundaries for safe operations

PC11. Explore the purpose and various types of operating procedures

Pre- Start-Up, start-up and shut-down Review

To be competent, the user/individual on the job must be able to:

PC12. Execute planned, unplanned, emergency, staged, and delayed start-up and shut-down activities as required.

PC13. Assess the system for new and modified facilities during commissioning.

PC14. Establish clear pre-startup procedures and adhere to OEM specifications

PC15. Keep documentation, including P&IDs, current and accurate.

PC16. Ensure that construction aligns with design specifications.

PC17. Obtain prior approval from the HSE department and verify safety procedures

PC18. Perform thorough hazard analysis before facility startup.

PC19. Modify facilities to meet management of change requirements.

Work Permit System

To be competent, the user/individual on the job must be able to:

PC20. Perceive the role, purpose, and features of the work permit system

PC21. Collect work permit from competent authority

PC22. Interact effectively with contractors during the process

PC23. Familiarize with different types of permits and their formats

PC24. Conduct Job Safety Analysis and Tool Box Talk

PC25. Confirm proper training on the work permit system

Bulk Storage Operations.

To be competent, the user/individual on the job must be able to:

PC26. Assess the risk associated with bulk storage operation

PC27. locate storage tanks and implement secure procedures for tank filling

PC28. Address possible hazards associated with floating roof and fixed roof tank safety, bunding, extreme weather protection, lightning strikes, and chemical warehousing

Electrical Safety, Fire Safety

To be competent, the user/individual on the job must be able to:

PC29. Identify and control electrical hazards

PC30. Implement measures to manage electrostatic risks

PC31. Adhere to electrical codes and industry-specific standards.

PC32. inspect and maintain safety gears regularly.

PC33. Identify potential points of ignition, categorize areas based on fire risk

PC34. Install effective fire systems as per lower and higher explosive limits

PC35. Handle risk associated with jet fires, pool fires, BLEVE, CVCE, and UVCE

PC36. Follow fire equipment compliance as per OISD standards

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PC37. Confirm uninterruptible power supply

Dangerous Substances.

To be competent, the user/individual on the job must be able to:

PC38. Properly store, label, and transport hazardous materials like Explosive, Oxidising, Flash point, Flammable items

PC39. Wear appropriate gear (PPE Kit like gloves, goggles, etc.) while using Track spill response and evacuation procedures

Reaction Hazards

To be competent, the user/individual on the job must be able to:

PC40. Evaluate potential chemical reactions like Exothermic reaction, Endothermic reaction, Thermal runaway reaction, effect of temp, pressure, and catalysts on rates of chemical reactions

PC41. Prevent accidental mixing

PC42. Confirm proper ventilation in reaction areas

PC43. Take proper protective measures while usage of reactive chemicals

Safety in using Steam

To be competent, the user/individual on the job must be able to:

PC44. Clearly mark steam and condensate lines

PC45. Use lockout/tagout and evaluate work areas before maintenance.

PC46. Wear appropriate heat-resistant gear

PC47. Ensure and wait for systems to cool to minimize burn risk.

PC48. Follow safety related to steam hazards and associated risk

Dust explosions

To be competent, the user/individual on the job must be able to:

PC49. Implement good housekeeping practices, regularly inspect and clean areas prone to dust accumulation.

PC50. Identify, Eliminate and control potential ignition sources (such as electrical equipment, sparks, or open flames).

PC51. Conduct systematic risk assessments (e.g., Dust Hazard Analysis - DHA) for specific materials and processes, evaluate risks associated with dust explosions and develop effective strategies.

PC52. Implement administrative and engineering controls to prevent dust explosions, ensure proper equipment maintenance and safety barrier upkeep.

PC53. Report hazards and incidents to supervisors related to dust accumulation for timely action

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understanding with industry-specific process safety standards and regulations.

KU2. Understanding of codes such as API 521 for pressure relief systems

KU3. Knowledge of international standards like ISO 14001 for environmental management

Qualification Pack

- KU4.** Ability to compile and maintain process safety information (PSI) for all relevant equipment and processes
- KU5.** Understanding of process flow diagrams (PFDs) and piping and instrumentation diagrams (P&IDs)
- KU6.** Knowledge of equipment design codes and specifications
- KU7.** Proficiency in conducting hazard identification studies such as HAZOP (Hazard and Operability Study) and HAZID (Hazard Identification).
- KU8.** Capability to perform risk assessments using methods like LOPA (Layer of Protection Analysis) and FMEA (Failure Modes and Effects Analysis)
- KU9.** Understanding of risk matrices and risk ranking systems.
- KU10.** Development and implementation of safe operating procedures (SOPs) for various processes and equipment.
- KU11.** Knowledge of process deviations and abnormal situations handling procedures
- KU12.** Conducting pre-start-up safety reviews (PSSR) to ensure equipment and processes are safe before commissioning
- KU13.** Understanding of start-up and shut-down procedures to minimize risks during operational transitions
- KU14.** Understanding and management of a permit-to-work system to control hazardous work activities
- KU15.** Knowledge of permit types such as hot work permits, confined space entry permits, and electrical isolation permits
- KU16.** Understanding of safe practices for storing hazardous materials in bulk storage tanks
- KU17.** Knowledge of fire prevention and protection measures, including fire detection and suppression systems
- KU18.** Understanding of chemical compatibility and segregation principles
- KU19.** Understanding of steam system safety principles, including steam trap maintenance and boiler operation

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** keep abreast by reading about new policies at an organization level
- GS2.** communicate effectively in writing
- GS3.** convey information effectively
- GS4.** discuss observations, new policies and schedules and activities
- GS5.** listen with full attention and comprehend the information given by various source about the site or equipment
- GS6.** use reasoning skills to identify and resolve problems
- GS7.** organize all process manuals so that sorting/ accessing information is easy
- GS8.** analyse information and use logic to address safety-related issues and problems.
- GS9.** determine the safety measures to be followed and how changes in conditions operations, and the environment will affect outcomes.
- GS10.** undertake and express new ideas and initiatives

Qualification Pack

GS11. Supervise periodically all the process safety systems

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Process Safety standards</i>	1	2	-	-
PC1. comprehend the Oil Industry Safety Directorate (OISD) standards pertaining to process safety, systems, equipment, and appliances in the oil and gas industry	1	2	-	-
<i>Process Safety Information.</i>	3	6	-	-
PC2. Review safety documents such as MSDS, Safety Manual, and Training records.	1	2	-	-
PC3. Examine design aspects including interlock systems, alarm systems, and electrical classification.	1	2	-	-
PC4. Check P&ID, SOPs, MOC, and SIMOPS.	1	2	-	-
<i>Hazard Identification & Risk Analysis (HIRA)</i>	5	10	-	-
PC5. Analyze risks and grasp the purpose and utility of risk analysis	1	2	-	-
PC6. Evaluate both quantitative and qualitative aspects in Quantitative Risk Assessment (QRA) studies	1	2	-	-
PC7. Conduct HAZOP studies to identify potential hazards	1	2	-	-
PC8. Implement strategies for effective barrier management	1	2	-	-
PC9. Learn concepts like HAZID, FMEA, and ALARP (As Low As Reasonably Practicable)	1	2	-	-
<i>Operating Procedures.</i>	2	4	-	-
PC10. Recognize the boundaries for safe operations	1	4	-	-
PC11. Explore the purpose and various types of operating procedures	1	2	-	-
<i>Pre- Start-Up, start-up and shut-down Review</i>	8	16	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Execute planned, unplanned, emergency, staged, and delayed start-up and shut-down activities as required.	1	2	-	-
PC13. Assess the system for new and modified facilities during commissioning.	1	2	-	-
PC14. Establish clear pre-startup procedures and adhere to OEM specifications	1	2	-	-
PC15. Keep documentation, including P&IDs, current and accurate.	1	2	-	-
PC16. Ensure that construction aligns with design specifications.	1	2	-	-
PC17. Obtain prior approval from the HSE department and verify safety procedures	1	2	-	-
PC18. Perform thorough hazard analysis before facility startup.	1	2	-	-
PC19. Modify facilities to meet management of change requirements.	1	2	-	-
<i>Work Permit System</i>	6	12	-	-
PC20. Perceive the role, purpose, and features of the work permit system	1	2	-	-
PC21. Collect work permit from competent authority	1	2	-	-
PC22. Interact effectively with contractors during the process	1	2	-	-
PC23. Familiarize with different types of permits and their formats	1	2	-	-
PC24. Conduct Job Safety Analysis and Tool Box Talk	1	2	-	-
PC25. Confirm proper training on the work permit system	1	2	-	-
<i>Bulk Storage Operations.</i>	3	6	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. Assess the risk associated with bulk storage operation	1	2	-	-
PC27. locate storage tanks and implement secure procedures for tank filling	1	2	-	-
PC28. Address possible hazards associated with floating roof and fixed roof tank safety, bunding, extreme weather protection, lightning strikes, and chemical warehousing	1	2	-	-
<i>Electrical Safety, Fire Safety</i>	10	17	-	-
PC29. Identify and control electrical hazards	2	1	-	-
PC30. Implement measures to manage electrostatic risks	1	2	-	-
PC31. Adhere to electrical codes and industry-specific standards.	1	2	-	-
PC32. inspect and maintain safety gears regularly.	1	2	-	-
PC33. Identify potential points of ignition, categorize areas based on fire risk	1	2	-	-
PC34. Install effective fire systems as per lower and higher explosive limits	1	2	-	-
PC35. Handle risk associated with jet fires, pool fires, BLEVE, CVCE, and UVCE	1	2	-	-
PC36. Follow fire equipment compliance as per OISD standards	1	2	-	-
PC37. Confirm uninterruptible power supply	1	2	-	-
<i>Dangerous Substances.</i>	2	4	-	-
PC38. Properly store, label, and transport hazardous materials like Explosive, Oxidising, Flash point, Flammable items	1	2	-	-
PC39. Wear appropriate gear (PPE Kit like gloves, goggles, etc.) while using Track spill response and evacuation procedures	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Reaction Hazards</i>	4	8	-	-
PC40. Evaluate potential chemical reactions like Exothermal reaction, Endothermic reaction, Thermal runaway reaction, effect of temp, pressure, and catalysts on rates of chemical reactions	1	2	-	-
PC41. Prevent accidental mixing	1	2	-	-
PC42. Confirm proper ventilation in reaction areas	1	2	-	-
PC43. Take proper protective measures while usage of reactive chemicals	1	2	-	-
<i>Safety in using Steam</i>	5	10	-	-
PC44. Clearly mark steam and condensate lines	1	2	-	-
PC45. Use lockout/tagout and evaluate work areas before maintenance.	1	2	-	-
PC46. Wear appropriate heat-resistant gear	1	2	-	-
PC47. Ensure and wait for systems to cool to minimize burn risk.	1	2	-	-
PC48. Follow safety related to steam hazards and associated risk	1	2	-	-
<i>Dust explosions</i>	6	10	-	-
PC49. Implement good housekeeping practices, regularly inspect and clean areas prone to dust accumulation.	1	2	-	-
PC50. Identify, Eliminate and control potential ignition sources (such as electrical equipment, sparks, or open flames).	2	2	-	-
PC51. Conduct systematic risk assessments (e.g., Dust Hazard Analysis - DHA) for specific materials and processes, evaluate risks associated with dust explosions and develop effective strategies.	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC52. Implement administrative and engineering controls to prevent dust explosions, ensure proper equipment maintenance and safety barrier upkeep.	1	2	-	-
PC53. Report hazards and incidents to supervisors related to dust accumulation for timely action	1	2	-	-
NOS Total	55	105	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N3608
NOS Name	Implement PSM practices within the organization
Sector	Hydrocarbon
Sub-Sector	
Occupation	Management of Health, Safety and Environment (HSE)
NSQF Level	5
Credits	2.5
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

HYC/N3609: Carry out PSM in oil and gas industries

Description

This unit covers basic practices that improve effectiveness of working in a team in an organizational set-up

Scope

The scope covers the following :

- This unit/task covers the following:
- Exploration and Production
- Refinery
- Pipeline Operations
- Marketing oil and gas product

Elements and Performance Criteria

Exploration and Production Refinery

To be competent, the user/individual on the job must be able to:

- PC1.** Ensure safe practices during drilling activities
- PC2.** Assess risks related to toxic, reactive, flammable, or explosive chemicals used in the operations
- PC3.** Implement safety protocols for production processes.
- PC4.** Quantify the level of risk posed by identified hazards
- PC5.** Create a system to respond to hazard assessment findings.
- PC6.** Safeguard workers during logging procedures
- PC7.** Engage employees and their representatives in hazard assessments and chemical accident prevention planning.
- PC8.** review the workplace hazard assessment and response system
- PC9.** Adapt to changes in processes and technology
- PC10.** Address safety concerns related to offshore logistics
- PC11.** Leverage real-time data and incident records for dynamic risk assessment
- PC12.** Establish safety guidelines for seismic surveys

Refinery

To be competent, the user/individual on the job must be able to:

- PC13.** Implement proper procedures for installing, using, and maintaining safety instruments and tools.
- PC14.** Regularly inspect and repair safety equipment to ensure its effectiveness in preventing accidents
- PC15.** Develop safety protocols during the design and construction phases of refinery operations
- PC16.** adhere to safety guidelines while operating and maintaining equipment
- PC17.** Establish safe practices for handling boilers

Qualification Pack

- PC18.** Ensure proper maintenance, inspection, and operation of boilers to prevent accidents or leaks
- PC19.** Create a robust emergency response plan
- PC20.** Train the personnel on effective responses during emergencies, including boiler-related incidents
- PC21.** review safety procedures and protocols.

Pipeline Operations

To be competent, the user/individual on the job must be able to:

- PC22.** Recognize the unique properties of petroleum and its derivatives being transported through pipelines.
- PC23.** Assess the factors like viscosity, volatility, and reactivity that impact safety during transportation
- PC24.** Design, Construct, Operate, and Maintain Pipelines as per OISD standards
- PC25.** Implement rigorous safety protocols during the design and construction phases
- PC26.** Adhere to guidelines for safe operation and regular maintenance of pipelines
- PC27.** Follow both statutory and company-specific safety regulations
- PC28.** Emphasize accident prevention through robust safety practices and risk assessments
- PC29.** Empower supervisors to oversee safety measures in pipeline operations
- PC30.** Monitor adherence to safety protocols, conduct inspections, and address any deviations
- PC31.** Identify potential hazards such as leaks, corrosion, or equipment failures
- PC32.** Implement preventive measures like leak detection systems, emergency shutdown (ESD) procedures, and regular integrity assessments

Marketing oil and gas product

To be competent, the user/individual on the job must be able to:

- PC33.** Identify the unique characteristics of each product (LPG, petrol, diesel, etc.)
- PC34.** Assess the hazards related to handling, storage, and transportation of petroleum product
- PC35.** Develop clear guidelines for safe handling, storage, and distribution
- PC36.** Address safety aspects during product transfer, loading, and unloading
- PC37.** Create comprehensive emergency response plans
- PC38.** Train personnel on swift and effective responses during incidents
- PC39.** Establish communication protocols for emergencies
- PC40.** Conduct routine inspections of storage facilities, pipelines, and transport vehicles
- PC41.** Maintain equipment integrity to prevent leaks, spills, or accidents
- PC42.** Monitor product quality to prevent contamination or degradation
- PC43.** inspect cylinders, tanks, and dispensing equipment regularly
- PC44.** Educate marketing personnel about product safety.
- PC45.** Foster a safety-conscious culture within the organization

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** Understanding of geological surveys and exploration techniques
- KU2.** Knowledge of drilling methods and equipment used in oil and gas extraction
- KU3.** Understanding of reservoir engineering principles and techniques
- KU4.** Familiarity with production operations, including well completion and maintenance
- KU5.** Knowledge of health, safety, and environmental regulations related to exploration and production activities
- KU6.** Understanding of refining processes and equipment used in converting crude oil into petroleum products
- KU7.** Knowledge of refinery safety protocols and procedures
- KU8.** Familiarity with quality control measures and testing methods in refinery operations
- KU9.** Understanding of pipeline construction, operation, and maintenance
- KU10.** Knowledge of pipeline safety regulations and standards
- KU11.** Familiarity with pipeline inspection and integrity management practices.
- KU12.** Understanding of transportation logistics and scheduling in pipeline operations
- KU13.** Knowledge of marketing strategies for oil and gas products.
- KU14.** Understanding of pricing mechanisms and market dynamics in the oil and gas industry
- KU15.** Familiarity with regulatory requirements for marketing and selling petroleum products
- KU16.** Knowledge of distribution networks and storage facilities for oil and gas products
- KU17.** Understanding of customer needs and preferences in the oil and gas market
- KU18.** Familiarity with risk management practices in oil and gas marketing activities
- KU19.** Knowledge of emerging trends and technologies in the oil and gas industry, impacting exploration, production, refining, pipeline operations, and marketing.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** keep abreast by reading about new policies at an organization level
- GS2.** communicate effectively in writing
- GS3.** convey information effectively
- GS4.** discuss observations, new policies and schedules and activities
- GS5.** listen with full attention and comprehend the information given by various source about the site or equipment
- GS6.** use reasoning skills to identify and resolve problems
- GS7.** organize all process manuals so that sorting/ accessing information is easy
- GS8.** analyse information and use logic to address safety-related issues and problems.
- GS9.** determine the safety measures to be followed and how changes in conditions operations, and the environment will affect outcomes.
- GS10.** undertake and express new ideas and initiatives
- GS11.** Supervise periodically all the process safety systems

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Exploration and Production Refinery</i>	12	24	-	-
PC1. Ensure safe practices during drilling activities	1	2	-	-
PC2. Assess risks related to toxic, reactive, flammable, or explosive chemicals used in the operations	1	2	-	-
PC3. Implement safety protocols for production processes.	1	2	-	-
PC4. Quantify the level of risk posed by identified hazards	1	2	-	-
PC5. Create a system to respond to hazard assessment findings.	1	2	-	-
PC6. Safeguard workers during logging procedures	1	2	-	-
PC7. Engage employees and their representatives in hazard assessments and chemical accident prevention planning.	1	2	-	-
PC8. review the workplace hazard assessment and response system	1	2	-	-
PC9. Adapt to changes in processes and technology	1	2	-	-
PC10. Address safety concerns related to offshore logistics	1	2	-	-
PC11. Leverage real-time data and incident records for dynamic risk assessment	1	2	-	-
PC12. Establish safety guidelines for seismic surveys	1	2	-	-
<i>Refinery</i>	9	18	-	-
PC13. Implement proper procedures for installing, using, and maintaining safety instruments and tools.	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Regularly inspect and repair safety equipment to ensure its effectiveness in preventing accidents	1	2	-	-
PC15. Develop safety protocols during the design and construction phases of refinery operations	1	2	-	-
PC16. adhere to safety guidelines while operating and maintaining equipment	1	2	-	-
PC17. Establish safe practices for handling boilers	1	2	-	-
PC18. Ensure proper maintenance, inspection, and operation of boilers to prevent accidents or leaks	1	2	-	-
PC19. Create a robust emergency response plan	1	2	-	-
PC20. Train the personnel on effective responses during emergencies, including boiler-related incidents	1	2	-	-
PC21. review safety procedures and protocols.	1	2	-	-
<i>Pipeline Operations</i>	11	22	-	-
PC22. Recognize the unique properties of petroleum and its derivatives being transported through pipelines.	1	2	-	-
PC23. Assess the factors like viscosity, volatility, and reactivity that impact safety during transportation	1	2	-	-
PC24. Design, Construct, Operate, and Maintain Pipelines as per OISD standards	1	2	-	-
PC25. Implement rigorous safety protocols during the design and construction phases	1	2	-	-
PC26. Adhere to guidelines for safe operation and regular maintenance of pipelines	1	2	-	-
PC27. Follow both statutory and company-specific safety regulations	1	2	-	-
PC28. Emphasize accident prevention through robust safety practices and risk assessments	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. Empower supervisors to oversee safety measures in pipeline operations	1	2	-	-
PC30. Monitor adherence to safety protocols, conduct inspections, and address any deviations	1	2	-	-
PC31. Identify potential hazards such as leaks, corrosion, or equipment failures	1	2	-	-
PC32. Implement preventive measures like leak detection systems, emergency shutdown (ESD) procedures, and regular integrity assessments	1	2	-	-
<i>Marketing oil and gas product</i>	13	26	-	-
PC33. Identify the unique characteristics of each product (LPG, petrol, diesel, etc.)	1	2	-	-
PC34. Assess the hazards related to handling, storage, and transportation of petroleum product	1	2	-	-
PC35. Develop clear guidelines for safe handling, storage, and distribution	1	2	-	-
PC36. Address safety aspects during product transfer, loading, and unloading	1	2	-	-
PC37. Create comprehensive emergency response plans	1	2	-	-
PC38. Train personnel on swift and effective responses during incidents	1	2	-	-
PC39. Establish communication protocols for emergencies	1	2	-	-
PC40. Conduct routine inspections of storage facilities, pipelines, and transport vehicles	1	2	-	-
PC41. Maintain equipment integrity to prevent leaks, spills, or accidents	1	2	-	-
PC42. Monitor product quality to prevent contamination or degradation	1	2	-	-
PC43. inspect cylinders, tanks, and dispensing equipment regularly	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC44. Educate marketing personnel about product safety.	1	2	-	-
PC45. Foster a safety-conscious culture within the organization	1	2	-	-
NOS Total	45	90	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N3609
NOS Name	Carry out PSM in oil and gas industries
Sector	Hydrocarbon
Sub-Sector	
Occupation	Management of Health, Safety and Environment (HSE)
NSQF Level	5
Credits	3.5
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

HYC/N9301: Working effectively in a team

Description

This unit covers basic practices that improve effectiveness of working in a team in an organizational set-up

Scope

The scope covers the following :

- Effective team work

Elements and Performance Criteria

Effective team work

To be competent, the user/individual on the job must be able to:

- PC1.** maintain clear communication with colleagues
- PC2.** pass on information to colleagues in line with organisational requirements
- PC3.** work in a team and support the team members
- PC4.** work in ways that show respect to colleagues
- PC5.** fulfil commitments made to colleagues
- PC6.** inform team members timely, if timelines can't be met
- PC7.** take the necessary initiatives to resolve the issues while working in team

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the organization policies and procedures related to team performance
- KU2.** the importance of effective communication and establishing good working relationships with colleagues
- KU3.** Deeper understanding of actions & consequences of gendered behaviour
- KU4.** the importance of creating an environment of trust and mutual respect
- KU5.** the implications of own work on the work and schedule of others
- KU6.** the standard practice in organization w.r.t communication at various levels
- KU7.** the instructions at individual level or at group level
- KU8.** individual work responsibility and corrective measures required to complete task in time
- KU9.** the importance of personal qualities like discipline and confidence to achieve success in work assigned

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.** communicate effectively in writing
- GS2.** read instructions, guidelines/procedures
- GS3.** communicate information effectively
- GS4.** make decisions on a suitable course of action or response keeping in view resource utilization while meeting commitments
- GS5.** how to plan the work to meet the deadline
- GS6.** the importance of consistent and reliable services for customer satisfaction
- GS7.** apply problem solving approaches in different situations
- GS8.** apply balanced judgments to different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Effective team work</i>	20	30	-	-
PC1. maintain clear communication with colleagues	2	4	-	-
PC2. pass on information to colleagues in line with organisational requirements	3	5	-	-
PC3. work in a team and support the team members	3	4	-	-
PC4. work in ways that show respect to colleagues	3	4	-	-
PC5. fulfil commitments made to colleagues	3	5	-	-
PC6. inform team members timely, if timelines can't be met	3	4	-	-
PC7. take the necessary initiatives to resolve the issues while working in team	3	4	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N9301
NOS Name	Working effectively in a team
Sector	Hydrocarbon
Sub-Sector	Generic
Occupation	Generic, Generic
NSQF Level	4
Credits	2.5
Version	5.0
Last Reviewed Date	27/08/2024
Next Review Date	26/08/2027
NSQC Clearance Date	27/08/2024

Qualification Pack

HYC/N9302: Maintain health, safety and security procedures

Description

This unit covers maintaining health, safety and security procedures at the workplace. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment

Scope

The scope covers the following :

- This unit/task covers the following:
- Practice health and safety measures
- Follow fire safety procedures
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Follow health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions
- PC2.** identify documents, location and people responsible for health and safety in the workplace
- PC3.** identify possible causes of risk or accident in the workplace
- PC4.** carry out safe working practices while dealing with hazards to ensure the safety of self and others
- PC5.** lift heavy objects safely using correct procedures
- PC6.** identify common safety signs, displayed in various areas

Follow fire safety procedures

To be competent, the user/individual on the job must be able to:

- PC7.** use the various appropriate fire extinguishers on different types of fires correctly
- PC8.** follow rescue techniques applied during fire hazard
- PC9.** follow good housekeeping practice in order to prevent fire hazards
- PC10.** list issues concerning the safety in work place
- PC11.** inform fire safety department about any near-miss incidents in the work place
- PC12.** follow the applicable laws, regulations and codes as per safety standard
- PC13.** prepare written accident/incident report and share with the concerned officer/department

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

- PC14.** provide appropriate first aid to victims in emergency situation
- PC15.** demonstrate basic techniques of bandaging
- PC16.** respond promptly and appropriately to an accident
- PC17.** perform rescue activity during an accident in real or simulated environments

Qualification Pack

PC18. demonstrate correct method to rescue injured people and others during an emergency

PC19. Demonstrate PPE policy (Job specific PPE)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company's policies on personnel management and duty reporting procedure
- KU2.** reporting structure within organization
- KU3.** how to escalate problem
- KU4.** concept of hazards and risks
- KU5.** health and safety hazards commonly affecting the work environment and related precautions
- KU6.** preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU7.** importance of using protective clothing/equipment while working
- KU8.** various causes of fire
- KU9.** techniques of using different types of fire extinguishers
- KU10.** different materials used for extinguishing fire
- KU11.** various types of safety signs and their significance

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate by writing
- GS2.** read instructions, guidelines/procedures and reports
- GS3.** communicate effectively and share the information efficiently
- GS4.** identify and report potential sources of danger
- GS5.** how to plan the work to meet the deadline
- GS6.** the importance of on time services
- GS7.** apply problem solving approaches in different situations
- GS8.** apply balanced judgments to different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow health and safety measures</i>	9	15	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions	1	4	-	-
PC2. identify documents, location and people responsible for health and safety in the workplace	2	1	-	-
PC3. identify possible causes of risk or accident in the workplace	1	3	-	-
PC4. carry out safe working practices while dealing with hazards to ensure the safety of self and others	2	2	-	-
PC5. lift heavy objects safely using correct procedures	1	3	-	-
PC6. identify common safety signs, displayed in various areas	2	2	-	-
<i>Follow fire safety procedures</i>	12	16	-	-
PC7. use the various appropriate fire extinguishers on different types of fires correctly	2	2	-	-
PC8. follow rescue techniques applied during fire hazard	2	2	-	-
PC9. follow good housekeeping practice in order to prevent fire hazards	2	2	-	-
PC10. list issues concerning the safety in work place	1	3	-	-
PC11. inform fire safety department about any near-miss incidents in the work place	2	2	-	-
PC12. follow the applicable laws, regulations and codes as per safety standard	1	3	-	-
PC13. prepare written accident/incident report and share with the concerned officer/department	2	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow emergencies, rescue and first-aid procedures</i>	4	19	-	-
PC14. provide appropriate first aid to victims in emergency situation	1	3	-	-
PC15. demonstrate basic techniques of bandaging	-	3	-	-
PC16. respond promptly and appropriately to an accident	-	3	-	-
PC17. perform rescue activity during an accident in real or simulated environments	1	4	-	-
PC18. demonstrate correct method to rescue injured people and others during an emergency	1	4	-	-
PC19. Demonstrate PPE policy (Job specific PPE)	1	2	-	-
NOS Total	25	50	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N9302
NOS Name	Maintain health, safety and security procedures
Sector	Hydrocarbon
Sub-Sector	Generic
Occupation	Generic, Generic
NSQF Level	4
Credits	2.5
Version	7.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training centre (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score a minimum of 70% of aggregate marks to successfully clear the assessment.

Qualification Pack

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 70

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HYC/N3606.Develop policies, SOPs and train to ensure safety at workplace	20	30	0	0	50	20
HYC/N3607.Incorporate the PSM system into the culture of leadership and organization	25	50	0	0	75	20
HYC/N3608.Implement PSM practices within the organization	55	105	0	0	160	20
HYC/N3609.Carry out PSM in oil and gas industries	45	90	0	0	135	10
HYC/N9301.Working effectively in a team	20	30	-	-	50	10
HYC/N9302.Maintain health, safety and security procedures	25	50	0	0	75	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	210	385	-	-	595	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
MIS	Management Information System
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
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Qualification Pack

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OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard operating procedure

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.